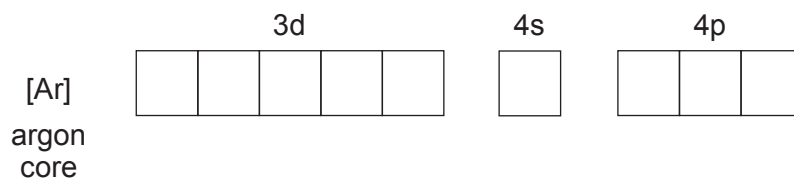


SECTION A

Answer **all** questions in the spaces provided.

1. Using the convention of arrows to represent electrons, complete the electronic structure of an arsenic atom, As. [1]



2. Give the oxidation number of molybdenum in the ion MoO_4^{2-} . [1]

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3. The half-life of the radioactive isotope oxygen-15 is 2 minutes.

- (a) Give the atomic number, mass number and symbol of the nuclide produced when 1 atom of ^{15}O decays by the emission of one positron. [1]

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- (b) If a quantity of ^{15}O decays by positron emission, determine what fraction of ^{15}O is left after 10 minutes. [1]

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4. Explain why the second molar ionisation energy of sodium is greater than its first molar ionisation energy. [1]

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